

Advanced Macroeconomics

Life Cycle Economies: Storesletten, Telmer & Yaron (2004, JME)

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Introduction

- **Goal:** introduce the incomplete markets framework in a OLG economy.
- Study a classic paper as an example: **Storesletten, Telmer & Yaron (2004): Consumption and risk sharing over the life cycle.**
- The life cycle structure is useful to study many questions where age interacts with inequality:
 - ▶ **Early age:** Education;
 - ▶ **Middle age:** Labor market;
 - ▶ **Old age:** Social security, health;

Storesletten, Telmer & Yaron (2004): Motivation

- Stylized Facts:

1. Inequality in consumption and earnings increase substantially during the life cycle;
2. The increase in inequality of consumption is less than earnings;
3. The increase is approximately linear.

- Can noninsurable idiosyncratic shocks to labor earnings explain these facts?
- What is the role of initial heterogeneity in comparison to earnings shocks during the life cycle?

Storesletten, Telmer & Yaron (2004): Method

- Estimate a rich labor earnings process using the PSID:
 - ▶ Individual fixed effects;
 - ▶ Persistent shocks;
 - ▶ Transitory shocks.
- Input the earnings process in an OLG model without consumption risk sharing.
 - ▶ General equilibrium pins down the level of wealth.
- Only two sources of insurance:
 - ▶ Self-insurance;
 - ▶ Pension system financed by labor tax.

Empirical Evidence

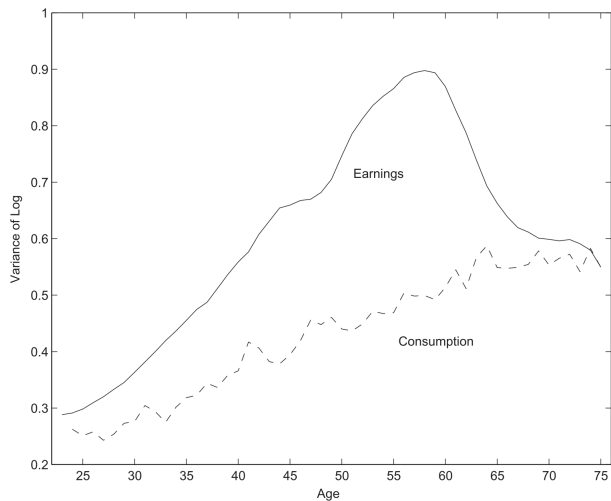
- Data

- ▶ **Panel Study of Income Dynamics (PSID, 1969-1992)**: Household survey; Panel data.
Earnings: wage income before taxes, plus transfers.
- ▶ **Consumption Expenditure Survey (CEX, 1980-1990)**: Consumption survey;
Consumption: nonmedical and nondurable expenditures on goods and services by urban U.S. households.

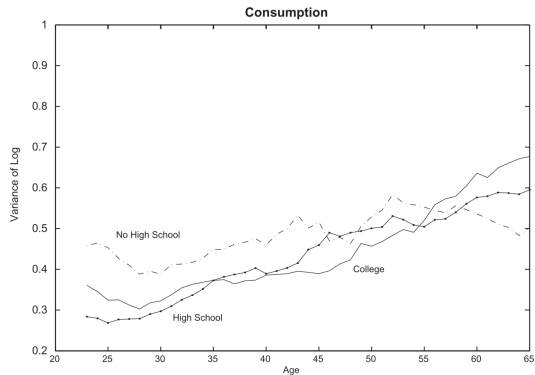
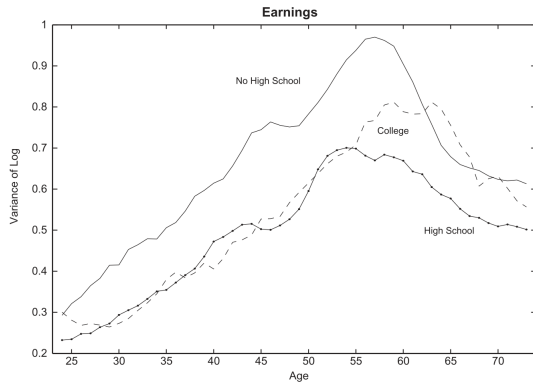
- Unit of study: household.

- Clean for cohort effects using a linear regression.

Empirical Evidence



Empirical Evidence: by Education



Earnings Stochastic Process

- For household i of age h , denote the **residual** log of annual earnings as u_{ih} (i.e., log income with mean zero and net of cohort effects).
- The stochastic process of u_{ih} is defined as:

$$u_{ih} = \alpha_i + \epsilon_{ih} + z_{ih}$$

$$z_{ih} = \rho z_{i,h-1} + \eta_{ih}$$

where $\alpha_i \sim N(0, \sigma_\alpha^2)$, $\epsilon_{ih} \sim N(0, \sigma_\epsilon^2)$, $\eta_{ih} \sim N(0, \sigma_\eta^2)$, and $z_{i0} = 0$.

- Interpretation of each idiosyncratic shock:
 - ▶ **Fixed effect**, α_i : Innate ability, early education investments, etc.
 - ▶ **Transitory shock**, ϵ_{ih} : Earnings bonus, transitory health problems, etc.
 - ▶ **Persistent shock**, η_{ih} : Unemployment shocks with scarring effects, promotions, etc.

Earnings Stochastic Process

- At $h = 0$ (age 22), the variance of u_{i0} is $V(u_{i0}) = \sigma_\alpha^2 + \sigma_\epsilon^2 + \sigma_\eta^2$.
- At $h = 1$ (age 23), the variance of u_{i1} is $V(u_{i1}) = \sigma_\alpha^2 + \sigma_\epsilon^2 + \sigma_\eta^2 + \rho^2 \sigma_\eta^2$.
- The variance of u_{ih} for age h :

$$V(u_{ih}) = \sigma_\alpha^2 + \sigma_\epsilon^2 + \sigma_\eta^2 \sum_{j=0}^{h-1} \rho^{2j}.$$

- The variance of earnings increases during the life-cycle as persistent shocks accumulate!
- The rate of the increase depends on how persistent are the shocks: ρ .
 - ▶ If $\rho = 1$, shocks are permanent and their effects never fade out.

Earnings Stochastic Process

- The goal is to estimate the parameters: ρ , σ_η^2 , σ_α^2 , and σ_ϵ^2 .
- Use GMM to estimate the parameters. Identification intuition (case $\rho = 1$). Take the difference

$$\Delta u_{ih} = \Delta \epsilon_{ih-1} + \eta_{ih}$$

and use the moments:

$$V(\Delta u_{ih}) = 2\sigma_\epsilon^2 + \sigma_\eta^2$$

$$COV(\Delta u_{ih}, \Delta u_{ih+1}) = -\sigma_\epsilon^2$$

- ▶ To recover σ_α^2 , use σ_η^2 , σ_ϵ^2 and the variance of levels, $V(u_{ih})$.
- ▶ To estimate ρ , an extra time period is required so we need a panel of at least 4 time periods.
- ▶ STY actually use all moments in levels. The broad idea is similar.

- The economy is populated by H overlapping generations. Denote ϕ_h as the unconditional probability of surviving up to age h , preferences are:

$$\mathbb{E} \sum_{t=1}^H \beta^h \phi_h \frac{c_h^{1-\gamma}}{1-\gamma}, \quad \text{where } \beta \in (0, 1).$$

- Agents begin to work at 22 and, conditional on surviving, retire at 65. At 100 die with certainty.
- Technology: $Y = ZK^\theta N^{1-\theta}$.
 - ▶ Firms hire labor and rent capital at prices W and R .
 - ▶ Law of motion: $K' = Y - C + (1 - \delta)K$.
 - ▶ The economy has SS growth rates of g , so some variables must be normalized.

Budget Constraint

- Budget constraint of a working agent:

$$c_h + (1 + g)a'_h \leq a_h R / \xi_h + n_h(1 - \tau)W$$

where τ is a labor tax, and $\xi_h = \phi_h / \phi_{h-1}$ is the survivor's premium.

- The labor endowment process is given by:

$$\log n_h = \kappa_h + u_h$$

where κ_h are the age-profile earnings common to all agents, while u_h is the individual-specific stochastic process as defined before.

Budget Constraint

- Budget constraint of a retired agent:

$$c_h + (1 + g)a'_h \leq a_h R / \xi_h + B(\bar{n}_h)W$$

where $B(\bar{n}_h)$ is the pension replacement rate that is a function of the average labor endowments over the life cycle, $\bar{n}_h$.

- The avg. labor endowment, $\bar{n}_h$, summarizes the social security contribution and evolves as following:

$$\bar{n}_{h+1} = \begin{cases} \bar{n}_h + n_h / I & \text{if working,} \\ \bar{n}_h & \text{if retired,} \end{cases}$$

where I is the number of years before retirement.

Value Function

- Let V_h denote the value function of an h -year-old agent. The value function of the agent is:

$$V_h(\alpha, z_h, \epsilon_h, a_h, \bar{n}_h) = \max_{a'_{h+1} \geq \underline{a}(\alpha, z, h)} \left\{ \frac{c_h^{1-\gamma}}{1-\gamma} + \hat{\beta} \xi_{h+1} \mathbb{E}_h[V(\alpha, z'_{h+1}, \epsilon'_{h+1}, a'_{h+1}, \bar{n}'_{h+1})] \right\}$$

s.t.

$$c_h + (1+g)a'_h = \begin{cases} a_h R / \xi_h + n_h(1-\tau)W & \text{if working,} \\ a_h R / \xi_h + B(\bar{n}_h)W & \text{if retired,} \end{cases}$$

where $\hat{\beta} = \beta(1+g)^{1-\gamma}$ and $\underline{a}(\alpha, z, h)$ is an age-dependent borrowing constraint.

- You can solve the value function using backward induction, as $V_{H+1} = 0$ and $a'_{H+1} = 0$.

Equilibrium

- Let $S = \{\alpha, z, \epsilon, a, \bar{n}, h\}$ be the state space.
- A stationary equilibrium is defined as prices, R and W ; a set of functions, $\{V_h, a'_{h+1}\}_{h=1}^H$; aggregate capital stock K and labor supply N ; and a cross-sectional distribution μ of agents across S , such that:
 - (a) Prices are given by the firm's marginal productivity of labor and capital;
 - (b) Functions $\{V_h, a'_{h+1}\}_{h=1}^H$ solve the individual's problem;
 - (c) Given individual decisions, the distribution μ is stationary;
 - (d) Pension tax satisfies the social security budget constraint: $\int_S B(\bar{n})d\mu = N(1 - \tau)$.
 - (e) Capital and labor market clears: $K = \int_S a_h d\mu$ and $N = \int_S n_h d\mu$.

- **Standard parameters:** $\theta = 0.4$, $\gamma = 2$, $\delta = 0.109$.
- **Stochastic process parameters:** $(\rho, \sigma_{\eta}^2, \sigma_{\epsilon}^2, \sigma_{\alpha}^2, \kappa_h)$ estimated using PSID.
- $B(\bar{n}_h)$ replicates the pension system in the US.
- $\beta = 0.961$ matches wealth-to-income ratio of 3.1 in the US.

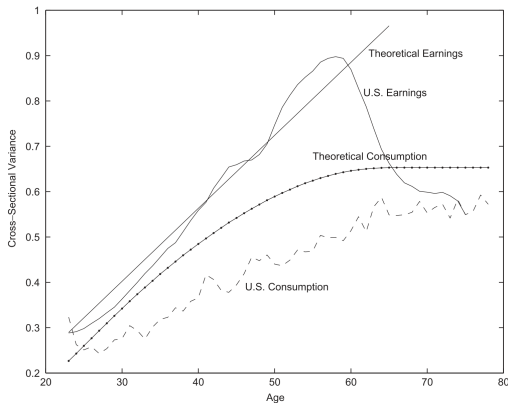
Results

Qualitatively Successful

- Consumption inequality is lower than earnings inequality;
- Earnings inequality increase faster than consumption inequality.

Quantitative: Consumption still a bit off.

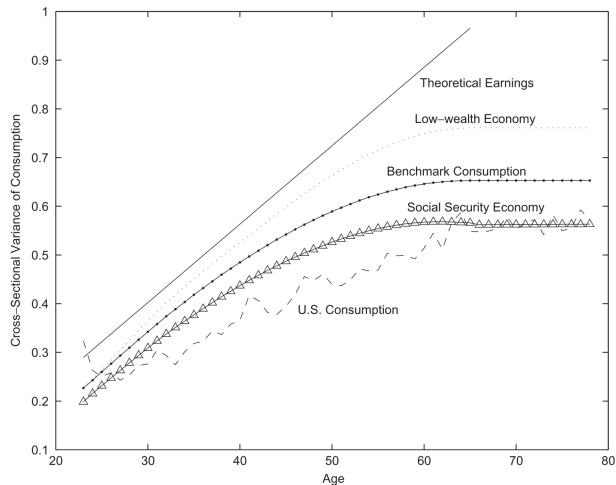
Figure: Model without Social Security ($B(\bar{n}_h) = 0$)



Results

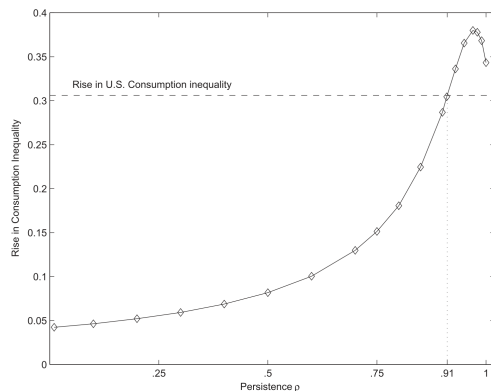
Social Security: it decreases consumption inequality, matches the data better;

Importance of Wealth: ↓ wealth-to-income ratio, ↓ the self-insurance and ↑ consumption inequality.



What matters for Consumption Inequality?

- To generate enough consumption inequality, we need shocks to have enough persistence.
- Borrowing constraints and initial wealth inequality: matters for inequality between 23-29, but it is not very important later.



Life-cycle shocks versus fixed effects

What type of inequality costs more for the agent?

- **Utilitarian measure:** how much consumption the agent is willing to forgo to live in a world without shocks?
- Let ψ the percentage consumption loss. Rewriting the utility function:

$$\mathbb{E} \sum_{t=1}^H \beta^h \phi_h \frac{[c_h(1 - \psi)]^{1-\gamma}}{1 - \gamma} = (1 - \psi)^{1-\gamma} \mathbb{E} \sum_{t=1}^H \beta^h \phi_h \frac{c_h^{1-\gamma}}{1 - \gamma} = (1 - \psi)^{1-\gamma} \mathbb{E} V_1(\alpha, z, \epsilon, 0),$$

where $\mathbb{E} V_1(\alpha, z, \epsilon, 0)$ is the average lifetime utility of a unborn agent (under the veil of ignorance).

- We can do the same thing for a model without risk, social security, etc.

Life-cycle shocks versus fixed effects

- Solve the model without risk and compute the expected VF at age 1: $\mathbb{E}\hat{V}_1(\alpha, 0|\text{no risk})$.
- What is ψ that equalizes expected utility in both worlds?

$$(1 - \psi)^{1-\gamma} \mathbb{E}\hat{V}_1(\alpha, 0|\text{no risk}) = \mathbb{E}V_1(\alpha, z, \epsilon, 0) \iff \psi = 1 - \left(\frac{\mathbb{E}V_1(\alpha, z, \epsilon, 0)}{\mathbb{E}\hat{V}_1(\alpha, 0|\text{no risk})} \right)^{1/(1-\gamma)}$$

- The **consumption equivalent variation** of each type of shock:
 - ▶ $\psi_{z,\epsilon} = 27.4\%$.
 - ▶ $\psi_{\alpha} = 20.2\%$.
- Shocks are costlier than ex-ante heterogeneity!

Conclusion

- Inequality in earnings and consumption increase during the life cycle.
- Persistent shocks are key to account for this regularity.
- Social security reduces welfare inequality.
- What other policies can achieve less welfare inequality?

Where to go now?

- **Pension System:** Conesa and Krueger (1999), Fuster et al (2007), McKiernan (2021).
- **Inequality over the Life cycle:** Huggett, Ventura and Yaron (2011), Guvenen, Kuruscu, Ozkan (2014).
- **Human Capital and Intergenerational Mobility:** Lochner and Monge-Naranjo (2011), Daruich (2020), Abbot et al (2019), Restuccia and Urrutia (2004).
- **Earnings Process:** De Nardi et al (2020), Guvenen et al (2021).
- **Welfare Policy:** Guner, Kaygusuz and Ventura (2021, WP); Low, Meghir and Pistaferri (2010, AER); Wellschmied (2021, QE).
- **Consumption Insurance:** Kaplan and Violante (2010), Blundell, Pistaferri and Preston (2008).
- **Marriage and Female Labor Supply:** Voena (2015), Attanasio, Low and Sanchez-Marcos (2008).
- **Old Age and Health Shocks:** many papers by Mariacristina Denardi and Eric French.